Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_random_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_190728_b
- started_at_local: 2026-05-13 19:07:28
- finished_at_local: 2026-05-13 19:10:23
- duration_hhmm: 00:03
- duration_seconds: 174.855
- seed_offset: 1000
- replicate_label: b
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.245937	0.00754	0.139983	0.838358	0.78	-0.013078
cvrp_random_replay	random	score_only	False	0.247013	0.005376	0.139619	0.914351	0.58	0.005856
cvrp_failure_replay	failure	score_only	False	0.252953	0.005376	0.142919	0.932697	0.58	0.052592
cvrp_failure_replay_compression	failure	novelty_gate	True	0.250364	0.005376	0.14148	0.921805	0.58	0.049644

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.245937, synthetic 0.00754, combined 0.139983.

- Accepted-epoch count 2, mean accepted code novelty 0.838358, and final complexity 0.78.
- Adaptation efficiency -0.013078 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.245937 across 5 instances; family means: A=0.078534, B=0.238022, E=0.303263, P=0.371841.
- Panel synthetic_holdout mean gap 0.00754 across 4 instances; family means: alternating_belt=0.03016, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 582, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.270742}, {"best_known_cost": 937, "cost": 1083, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.155816}, {"best_known_cost": 784, "cost": 882, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.125}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.247013, synthetic 0.005376, combined 0.139619.
- Accepted-epoch count 3, mean accepted code novelty 0.914351, and final complexity 0.58.
- Adaptation efficiency 0.005856 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.247013 across 5 instances; family means: A=0.115183, B=0.241387, E=0.332054, P=0.305054.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1065, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.136606}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.252953, synthetic 0.005376, combined 0.142919.
- Accepted-epoch count 2, mean accepted code novelty 0.932697, and final complexity 0.58.
- Adaptation efficiency 0.052592 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.252953 across 5 instances; family means: A=0.123037, B=0.260946, E=0.314779, P=0.305054.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.145144}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.

- Final transfer gaps: CVRPLIB 0.250364, synthetic 0.005376, combined 0.14148.
- Accepted-epoch count 2, mean accepted code novelty 0.921805, and final complexity 0.58.
- Adaptation efficiency 0.049644 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.250364 across 5 instances; family means: A=0.123037, B=0.254873, E=0.301344, P=0.31769.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 585, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.277293}, {"best_known_cost": 937, "cost": 1072, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.144077}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of CVRP Replay-Aware Benchmark Results

Condition	Replay Mode	Final CVRPLIB Gap	Synthetic Holdout Gap	Transfer Gap	Adaptation Efficiency	Complexity	Code Novelty (mean)	Notes
cvrp_no_replay	None	**0.2459**	0.00754	0.1400	-0.0131	0.78	0.838	Best held-out CVRPLIB gap. Higher complexity.
cvrp_random_replay	Random Replay	0.2470	**0.00538**	**0.1396**	+0.00586	0.58	**0.914**	Best synthetic & transfer gaps. Code novelty ↑, complexity ↓.
cvrp_failure_replay	Failure Replay	0.2530	0.00538	0.1429	**+0.0526**	0.58	0.933	Worst CVRPLIB gap but best adaptation efficiency. High novelty.
cvrp_failure_replay_compression	Failure Replay + Compression	0.2504	0.00538	0.1415	+0.0496	0.58	0.922	Near failure replay performance, with compression pressure.

Interpretation

- **Transfer Performance:**
- The best transfer metrics (held-out CVRPLIB gap, synthetic gap, and mean transfer gap) are overall achieved by the **random replay** condition.
- Synthetic holdout gap is lowest (0.00538) here.
- Transfer gap (~0.1396) is slightly better than no replay (0.1400).
- No replay achieves the best held-out CVRPLIB gap (0.2459) but slightly worse synthetic gap and similar transfer gap.
- Failure replay conditions show marginally worse held-out performance (CVRPLIB gap ~0.25) but slightly higher transfer gap, with better adaptation efficiency.

- **Code Novelty vs Transfer:**
- Code novelty is highest in failure replay scenarios (~0.93), while random replay also has increased code novelty (~0.91) compared to no replay (~0.84).
- Despite higher code novelty in failure replay, this does not translate to better CVRPLIB or synthetic gaps compared to random replay.
- Random replay condition improves transfer metrics while code novelty also increases and complexity decreases relative to no replay.
- Hence, increases in code novelty under failure replay are not accompanied by better transfer performance, contrary to random replay where code novelty gain aligns with improved transfer.

- **Replay Mode Effects:**
- **No Replay (baseline):** Achieves best held-out CVRPLIB gap but worse synthetic gap and transfer gap.
- **Random Replay:** Best synthetic and transfer gaps with modest CVRPLIB gap, lower complexity, and higher code novelty. Suggests replay of random past cases improves generalization and transfer more than no replay.
- **Failure Replay:** Slightly worse CVRPLIB and transfer gaps than random replay, but higher adaptation efficiency and code novelty. Indicates focused replay of failure cases boosts adaptation but not best overall transfer.
- **Failure Replay + Compression:** Similar trends as failure replay without compression but with compression pressure and novelty-based selection mode.

Conservative Conclusions

- The **random replay** condition offers the best combination of transfer performance (lowest synthetic and transfer gaps) and code novelty with reduced algorithmic complexity relative to no replay.
- Although **no replay** attains a marginally better CVRPLIB held-out gap, its synthetic and transfer performance are inferior to random replay.
- **Failure replay** variants increase code novelty and adaptation efficiency but do not improve held-out CVRPLIB or synthetic gaps over random replay, indicating limited benefit for transfer.
- The data indicates that increased code novelty alone (especially under failure replay) is not sufficient for better transfer if not supported by deterministic gap metrics.
- Compression-aware failure replay does not improve performance over failure replay without compression.
- Lexical/code novelty changes should not be interpreted as algorithmic innovation without consistent improvements in optimality gaps and transfer metrics.

Recommendation

For better transfer generalization on CVRP benchmarks, especially in real-world held-out sets and synthetic distributions:

- Prioritize **random replay** over no replay and failure replay approaches.
- Treat increases in code novelty without concurrent optimality gain cautiously.
- Replaying diverse past random cases appears more effective than focusing on failure cases alone.